

TUSHER MONDAL

Bengaluru - 560001, Karnataka, India

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Adept at learning new technologies, I bring a diverse skillset in Data Analytics, Web Development, and Software Development to build scalable solutions.

Experience

HSBC

Present

Manager - Management Reporting - CIB

Bengaluru

- Own management reporting for the Corporate & Investment Banking (CIB) division, turning raw numbers into actionable insights for enterprise stakeholders.
- Work across the analytics stack with Microsoft Excel (including VBA), Power BI, TM1 and Alteryx.
- Automate repetitive tasks by building custom internal tools, streamlining workflows and maximizing time efficiency and productivity.
- **Technologies** : Excel, VBA, Power BI, Power Query, TM1, Alteryx, SQL, Python

HSBC

Jan 2024 – Jul 2024

Analyst Intern (converted to full-time)

Chennai

- Built a system that automates the generation of email reports that monitor data quality across specified businesses and regions. The reports are comprehensive and customized, delivered to a designated list of recipients.
- Automated repetitive tasks by building custom internal tools which results in streamlined workflows and maximized time efficiency and productivity.
- Uncovered Business Trends Through Data Exploration and Visualization.
- **Technologies** : Python/ (pandas, openpyxl, xlwings, pywin32 etc.), Power Query, Microsoft Excel, VS Code, GitHub

Celebal Technologies

Jun 2023 – Aug 2023

Data Science Intern

Remote

- Created a forecasting model that utilizes user traits to track behavior and predict future bandwidth consumption.
- Developed an interactive full stack panel that visualizes usage report on various metrics.
- **Technologies** : Python/ (pandas, scikit-learn, etc.), VS Code, Data Space, No-SQL

CodeClause

May 2023 – Jun 2023

Python Developer Intern

Remote

- Created a Web-Based URL Shorten-er which enables share large URLs in short form using Data Base mapping.
- Engineered a Voice Assistant CLI that leverages Google's Gemini(Previously Bard) model to mimic a conversation.
- **Technologies** : Python/ (Flask, speechrecognition, Bard-API, pyttsx3), VS Code, GitHub, MongoDB,

Projects

Writer | *Python, LLM, SDXL, TTS*

- A local AI agent that writes complete stories – a 3-stage pipeline that drafts the story with a local LLM, generates photorealistic book covers with RealVisXL (SDXL), and narrates it as a dual-voice audiobook with Orpheus TTS.
- Strict-JSON outline generation with schema-aware retries, embedded image backend with CPU offload, and an A5 PDF builder.

Piston-And-Sound | *JavaScript, Web Audio API*

- A web interface that mimics the physics-based sound of different engine types – procedural audio synthesized in-browser from engine physics: cylinder count, RPM, firing order and attack shapes.

Rusty Radio | *Rust*

- An intentionally simple music player that consumes bare-minimum system resources. Old-school, limited-feature tech that just works in the background.

HandMouse | *Python, OpenCV*

- Turns your hand gestures into mouse actions using just a webcam – computer vision meets everyday utility.

Socket Transfer | *Python, socket programming, TCP/IP*

- Facilitated a server-client system that can transfer data/files within an inter-network environment while utilizing the full capacity of the NIC.
- Executed automatic chunk signing technique that helps in faster rearranging resulting efficient transfer.

Railway Reservation System | *Python, Flask, HTML/CSS, Bootstrap, JavaScript, MongoDB*

- Formulated a Train Booking System with multi-level access design allowing users to book seats of their choice in a upcoming reservation schedule, similarly admins can manage upcoming reservations, train activity, and transactions.

Education

Jadavpur University

Master of Computer Application : 8.52 CGPA [Lin. Avg.]

Sep 2022 – Jul 2024

Jadavpur, West Bengal

West Bengal State University (Acharya Prafulla Chandra College)

Bachelor of Science in Computer Science Honors : 9.4 CGPA

Jul. 2019 – Aug 2022

Barasat, West Bengal

Co-curricular Activities

Final year thesis: Customizing **MulVAL** for Attack Graph Generation. Network Simulated Digital Forensic.

Certifications: Achieved HackerRank certification on Python, Java, and SQL

Competitive Programming: Solved more than 200 problems listed on **Leetcode, HackerRank** etc.

Technical Skills

Languages: Python, C, C++, Java, HTML/CSS, JavaScript, SQL, Bash, Rust

Dev Tools/Technologies: VS Code, Postman, GitHub, Flask, React, Microsoft Excel, Power BI, Alteryx, TM1, Nmap, Nessus, Oval

Databases: MongoDB, Redis, Data Space, MySQL, Microsoft Access

Local AI: LM Studio, Local LLMs, Stable Diffusion (SDXL / RealVisXL), Orpheus TTS

Relevant Courses: Computer Networks, Machine Learning, Advanced Programming in Python and Java. etc.